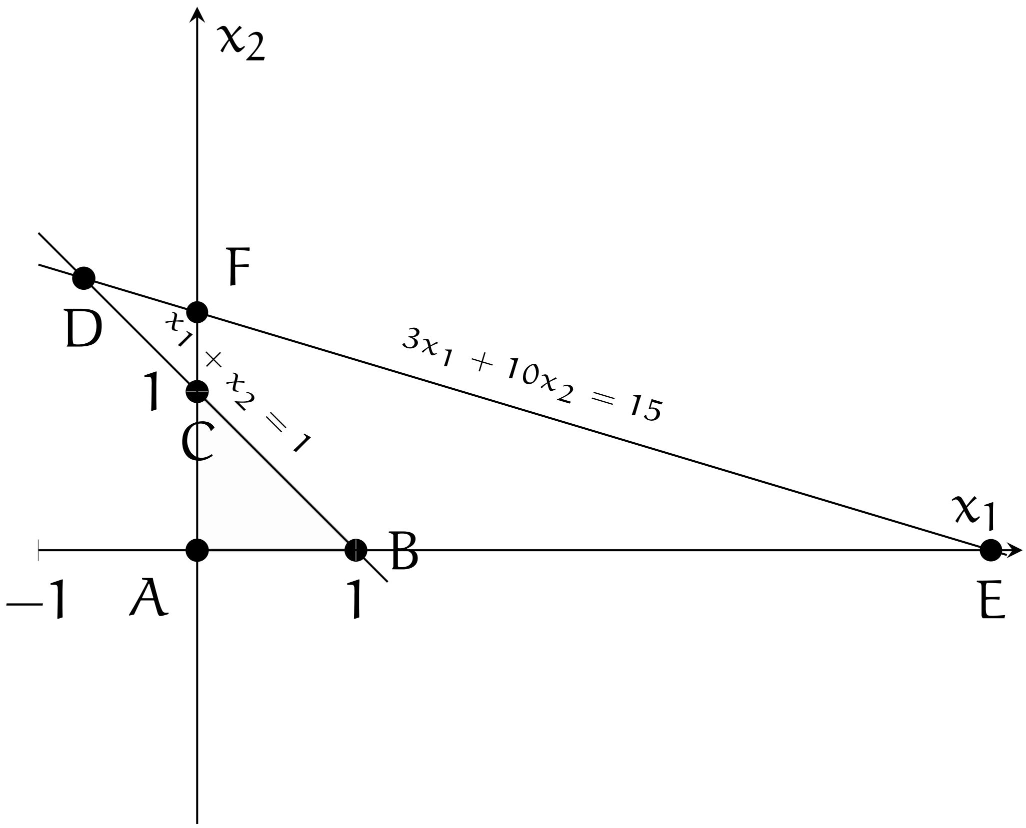
We represent the polyhedron

$$P = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 | x_1 + x_2 \leq 1, 3x_1 + 10x_2 \leq 15, x_1 \geq 0, x_2 \geq 0 \right\}$$

in the following figure:



The corresponding polyhedron in standard form is

$$Q = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4 | Ax = b, x \geq 0 \right\},$$

with

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 3 & 10 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 15 \end{pmatrix},$$

with the variables x_3 and x_4 being slack variables. Each basic solution is obtained by selecting 2 variables out of 4 to be in the basis. There is a total of 6 possible selections of basic variables.
The matrix $B = (A_{j1}, ..., A_{jm})$ is composed of columns $j_1, ..., j_m$ of the matrix A.

1. Basic solution with x_1 and x_2 in the basis ($j_1 = 1$,$j_2 = 2$).

$$B = \begin{pmatrix} 1 & 1 \\ 3 & 10 \end{pmatrix}; B^{-1} = \begin{pmatrix} \frac{10}{7} & -\frac{1}{7} \\ -\frac{3}{7} & \frac{1}{7} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} -\frac{5}{7} \\ \frac{12}{7} \end{pmatrix}; x = \begin{pmatrix} -\frac{5}{7} \\ \frac{12}{7} \\ 0 \\ 0 \end{pmatrix}.$$

This basic solution is not feasible because $B^{-1}b \not\geq 0$. It corresponds to point $D = (-\frac{5}{7}, \frac{12}{7})$ in the figure.

2. Basic solution with x_1 and x_3 in the basis $(j_1 = 1, j_2 = 3)$.

$$B = \begin{pmatrix} 1 & 1 \\ 3 & 0 \end{pmatrix}; B^{-1} = \begin{pmatrix} 0 & \frac{1}{3} \\ 1 & -\frac{1}{3} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 5 \\ -4 \end{pmatrix}; x = \begin{pmatrix} 5 \\ 0 \\ -4 \\ 0 \end{pmatrix}.$$

This basic solution is not feasible because $B^{-1}b \not\geq 0$. It corresponds to point $E = (5, 0)$.

3. Basic solution with x_1 and x_4 in the basis $(j_1 = 1, j_2 = 4)$.

$$B = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 12 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 12 \end{pmatrix}.$$

This basic solution is feasible and corresponds to point $B = (1, 0)$.

4. Basic solution with x_2 and x_3 in the basis $(j_1 = 2, j_2 = 3)$.

$$B = \begin{pmatrix} 1 & 1 \\ 10 & 0 \end{pmatrix}; B^{-1} = \begin{pmatrix} 0 & \frac{1}{10} \\ 1 & -\frac{1}{10} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix}; x = \begin{pmatrix} 0 \\ \frac{3}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix}.$$

This basic solution is not feasible because $B^{-1}b \not\geq 0$. It corresponds to point $F = (0, \frac{3}{2})$ in the figure.

5. Basic solution with x_2 and x_4 in the basis $(j_1 = 2, j_2 = 4)$.

$$B = \begin{pmatrix} 1 & 0 \\ 10 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ -10 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 5 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 5 \end{pmatrix}.$$

This basic solution is feasible and corresponds to point $C = (0, 1)$ in the figure.

6. Basic solution with x_3 and x_4 in the basis $(j_1 = 3, j_2 = 4)$.

$$B = B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 15 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 15 \end{pmatrix}.$$

This basic solution is feasible and corresponds to point $A = (0, 0)$ in the figure.

In summary, out of the six basic solutions, only three are feasible and correspond to vertices of the constraint polyhedron.